

Python Assignment: Data Structures, Functions, Numpy & Pandas

Instructions

- Attempt all questions.
- Submit the completed assignment in .ipynb format.
- Use appropriate function definitions and comments for clarity.

Assignment Questions

Section A: Data Structures & Control Structures

1. Create a list of 5 integers. Perform and print the result of the following operations: **append**, **extend**, **insert**, **remove**, **pop**, **clear**, **index**, **count**, **sort**, and **reverse**.
2. Create a tuple that stores 3 student names. Try changing the second name in the tuple. What happens? Explain why.
3. Create a set of integers with some duplicate values. Print the set and explain the output.
4. Create a dictionary with the keys: 'name', 'age', and 'city'. Update the age and add a new key 'email'. Print the final dictionary.
5. Write a script that checks if a person is eligible to vote ($\text{age} \geq 18$). Take age as a variable and print the appropriate message.
6. Given a 'marks' variable, print the grade:
 - 90 and above: 'A'
 - 75-89: 'B'
 - 50-74: 'C'
 - Below 50: 'Fail'
7. Given a number, check if it's positive, and if it is also even. If not positive, print if it's zero or negative.

Section B: Numpy

8. Create:
 - A scalar using `np.array(5)`
 - A 1D array with values 1 to 5
 - A 2D array (2x3) with values from 10 to 60 in steps of 10
9. Generate a 4x4 Numpy array of random integers between 0 and 100 using `np.random.randint()`.
10. Create a 2D numpy array of shape (3x3). Convert it into a pandas DataFrame and add column names: 'A', 'B', 'C'.

Section C: Exploring Pandas

11. Create a small DataFrame manually with 10 rows and columns: 'Name', 'Age', 'City', and 'Salary'. Then:

- Use `.info()`, `.describe()`
- Select 'Name' and 'City' columns
- Drop 'City' column
- Fill any missing values in 'Salary' column with the mean
- Remove any duplicate rows